dirac16complex: provenance

Sources, decisions, derivations and the verification record of the Dirac equation and the Kohn–Sham theories of the 16-component field dirac16complex — sections 7–10 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, what was used (the first part of 7+1.nb and the literature), every decision taken with the equation behind it, how each result was derived, and how it was checked: the 74 checks of sections 7–9 of 7+1.nb with their results, the 90 boolean checks of the independent Python program, the comparison of every printed number, the program versions and the SHA-256 hashes of the files.

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1. The task

This stage of the project asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Back up 7+1.nb, then append to it (the first part of the notebook stays as it is).
2. A valid and checked Dirac equation on the real 8-dimensional spacetime with one time and seven space dimensions, for a 16-component quantized fermion field named dirac16complex whose components are complex-valued Grassmann (anticommuting) operators.
3. Eight 16×16 Dirac matrices taken from an appropriate set of anticommuting real matrices of 7+1.nb; they MUST be purely real.
4. Chiral projections that write the 16-component field equation as two 8-component field equations, coupled when the mass is not zero.
5. A 16-component Kohn–Sham theory of dirac16complex coupled to itself.
6. A chiral version of that Kohn–Sham theory, using the chiral projections.
7. Complete, precise and checked documentation; for every documentation file a companion provenance file and a symbol file, each in Markdown and LaTeX, the LaTeX compiled to PDF; no document may refer the reader to another document, and every equation used must be written in the document that uses it.
8. Push every completed stage to the repository github.com/once-ere/7plus1 (the repository was renamed from 7+1 to 7plus1 during this stage), respecting the repository's .gitignore.
9. Use only the Wolfram front end WolframNB.exe of Wolfram 15.0.1, installed in `C:\Program Files\Wolfram Research\Wolfram\15.0.1`.

The coordinates are those of 7+1.nb and were not renamed: $X = (x_0, x_1, \ldots, x_7)$ with x_0 the ordinary time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary 3-space and $x_4, \ldots, x_7$ four auxiliary space-like dimensions. How each requirement was met:

#	how it was met	where in 7+1.nb
1	the evaluated notebook of the first stage was copied to backups/7+1_2026-10-26_before_dirac16complex.nb (section 2.1); sections 0–6 of the new notebook are cell for cell identical to it (section 4.3)	sections 0–6 unchanged
2	$(i\Gamma^A\partial_A - m)\psi = 0$ for `dirac16complex = Table[ψ16[a] @@ X, {a, 1, 16}]`; Klein–Gordon, Hamiltonian, plane waves, Grassmann algebra, mode expansion, canonical anticommutators, charge conjugation and Lorentz covariance (section 3.2–3.6)	7.1–7.9
3	$\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$: the real generators `Γ17[A]` of the first stage; every entry 0 or ±1 (section 3.1)	7.1
4	complex chirality $\chi = i\Gamma_0 \cdots \Gamma_7$ (no real one exists), $P_{L,R} = \frac{1}{2}(1 \mp \chi)$, and the two coupled equations $(i\partial_0 \pm \tau_i^O \partial_i)\phi_{L,R} = m\,\phi_{R,L}$ (section 3.7)	7.10–7.11
5	self-coupling through scalar, pseudoscalar, vector and axial-vector contact interactions; Hohenberg–Kohn theorem; Kohn–Sham equations; exchange (Wick), Fierz; uniform matter (sections 3.8–3.11, 3.13)	8.1–8.7
6	chiral densities κ, j_L, j_R; complex Kohn–Sham mass; 8+8-component Kohn–Sham equations; chiral symmetry; the chiral vacuum with the Dirac sea (sections 3.12, 3.14)	9.1–9.6
7	the documentation of this stage (a main document, a symbol dictionary and this record, each .md/.tex/.pdf), generated by tools/make_docs.py (section 5)	—
8	commits pushed to github.com/once-ere/7plus1 by the Wolfram job `tools/git_push.wl` (section 5)	—
9	every evaluation of 7+1.nb was made by that front end (section 4.1)	—

2. Inputs

2.1 The notebook 7+1.nb of the first stage

Before anything was appended, the evaluated notebook of the first stage (128 checks, all True) was copied, in the workspace `D:\Developer\github\7+1`, to the file below. The folder backups/ is excluded from the repository by its .gitignore, so the copy stays in the workspace only.

```
backups/7+1_2026-10-26_before_dirac16complex.nb
306963 bytes
SHA-256 f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
```

Sections 7–10 use the following objects of sections 1–5 of 7+1.nb, unchanged (Mathematica names in typewriter). Coordinates, metric and derivative operators:

$X = (x_0, x_1, \ldots, x_7)$ (`X`), $\quad \eta = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1)$ (`η17`), $\quad \partial_A = \partial/\partial x_A$,
$\Gamma^A \partial_A \psi$ (`DiracOp17`), $\quad \Box\psi = \eta^{AB}\partial_A\partial_B\psi = \partial_0^2\psi - \partial_1^2\psi - \cdots - \partial_7^2\psi$ (`waveOp17`).

The real 8×8 octonion matrices `τO[i]` ($\tau_0^O = 1_8$), obtained in the first stage from the split-octonion matrices τ_A of 4+4.nb as $\tau_i^O = \tau_i$ $(i = 1,2,3)$ and $\tau_k^O = \tau_k\,\sigma$ $(k = 4,\ldots,7)$, $\sigma = \tau_1\tau_2\tau_3 = \left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$:

$$\tau_1^O = \begin{pmatrix} 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_2^O = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_3^O = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_4^O = \begin{pmatrix} 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0 \end{pmatrix},$$

$$\tau_5^O = \begin{pmatrix} 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0 \end{pmatrix}, \quad \tau_6^O = \begin{pmatrix} 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0 \end{pmatrix},$$

$$\tau_7^O = \begin{pmatrix} 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0 \end{pmatrix};$$

the split matrices used for them are

$$\tau_4 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_6 = \begin{pmatrix} 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1 \end{pmatrix},$$

and $\tau_i = \tau_i^O$ for $i = 1, 2, 3$. They satisfy

$$(\tau_i^O)^{\mathsf{T}} = -\tau_i^O, \qquad \tau_i^O \tau_j^O + \tau_j^O \tau_i^O = -2\,\delta_{ij}\,1_8, \qquad \tau_1^O \cdots \tau_7^O = 1_8.$$

The real 16×16 generators `Γ17[A]` of the Clifford algebra of 7+1 spacetime, their products and the objects built from them:

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \Gamma^A = \eta^{AA}\Gamma_A \ (\texttt{Γ17up[A]}),$$

$$\Gamma_8 \equiv \omega = \Gamma_0\Gamma_1\cdots\Gamma_7 \ (\texttt{Γ17[8]}), \qquad \chi = i\,\omega \ (\texttt{χ17}), \qquad P_{L,R} = \tfrac{1}{2}(1_{16} \mp \chi) \ (\texttt{PL17},\ \texttt{PR17}),$$

$$\Sigma_{AB} = \tfrac{1}{4}(\Gamma_A\Gamma_B - \Gamma_B\Gamma_A) \ (\texttt{Σ17[A, B]}), \qquad \Gamma_K = \Gamma_{A_1}\cdots\Gamma_{A_k} \ (K = \{A_1 < \cdots < A_k\};\ \texttt{products256}),$$

$$\mathcal{B}(\beta_{\mathrm{b}}) = \cosh\tfrac{\beta_{\mathrm{b}}}{2}\,1_{16} + \sinh\tfrac{\beta_{\mathrm{b}}}{2}\,\Gamma_0\Gamma_1 \ (\texttt{Sβ}), \qquad L_{BK} = \tfrac{1}{16}\mathrm{tr}\big(\eta_{BB}\Gamma_B\,\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1}\big) \ (\texttt{Λboost}).$$

In terms of the 16×16 Dirac matrices of 4+4.nb, $T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$ with $\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\tau_A^{\mathsf{T}}\sigma$ and $T_AT_B + T_BT_A = 2\eta_{AB}^{(4,4)}1_{16}$, $\eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$, the first stage obtained $\Gamma_0 = T_0$, $\Gamma_i = T_0T_i$ $(i = 1, 2, 3)$ and $\Gamma_k = T_0T_kT_4T_5T_6T_7$ $(k = 4, \ldots, 7)$.

2.2 Literature, and the equation taken from each source

- P. A. M. Dirac, Proc. R. Soc. Lond. A 117 (1928) 610 — the first-order wave equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ with $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$; here written in 7+1 dimensions with the eight Γ^A.
- P. Hohenberg and W. Kohn, Phys. Rev. 136 (1964) B864 — for a non-degenerate ground state the density determines the external potential up to a constant, and $E[v] = \min_n \big(F[n] + \int v\,n\big)$; the proof by the two variational inequalities is used in section 3.10.
- W. Kohn and L. J. Sham, Phys. Rev. 140 (1965) A1133 — a non-interacting reference system with the same density: $\big[-\tfrac{1}{2}\nabla^2 + v + v_H + v_{xc}\big]\varphi_k = \varepsilon_k\varphi_k$, $n = \sum_k |\varphi_k|^2$, $v_{xc} = \delta E_{xc}/\delta n$ (atomic units); here its relativistic form.
- A. K. Rajagopal and J. Callaway, Phys. Rev. B 7 (1973) 1912; A. H. MacDonald and S. H. Vosko, J. Phys. C 12 (1979) 2977 — relativistic density-functional theory: the energy is a functional of the current j^A, and the Kohn–Sham equations have the Dirac form $\big[\alpha_i(-i\partial_i + \mathcal{V}_i) + \beta m + \mathcal{V}_0\big]\varphi_k = \varepsilon_k\varphi_k$ with the effective potential $\mathcal{V}_A = \delta(E - T_s)/\delta j^A$.
- M. Levy, Proc. Natl. Acad. Sci. USA 76 (1979) 6062; E. H. Lieb, Int. J. Quantum Chem. 24 (1983) 243 — the universal functional by constrained search, $F[n] = \min_{\Psi\to n}\langle\Psi|\hat T + \hat W|\Psi\rangle$, and Lieb's form $F[n] = \sup_v \big(E[v] - \int v\,n\big)$.

- R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990), ch. 8; E. Engel and R. M. Dreizler, *Density Functional Theory: An Advanced Course*, Springer (2011) — the no-sea approximation: the densities are sums over the occupied positive-energy Kohn–Sham states only, $j^A = \sum_{0<\varepsilon_k\le\varepsilon_F} \bar\varphi_k \Gamma^A \varphi_k$.
- J. D. Walecka, Ann. Phys. 83 (1974) 491; B. D. Serot and J. D. Walecka, Adv. Nucl. Phys. 16 (1986) 1 — relativistic mean field of uniform matter: $m^* = m - \frac{g_\sigma^2}{m_\sigma^2}\rho_S$, $V_0 = \frac{g_\omega^2}{m_\omega^2}\rho$, $\varepsilon = \frac{g_\omega^2}{2m_\omega^2}\rho^2 + \frac{m_\sigma^2}{2g_\sigma^2}(m - m^*)^2 + \varepsilon_{\rm kin}(m^*)$; the contact couplings here correspond to $g_S = -g_\sigma^2/m_\sigma^2$ and $g_V = g_\omega^2/m_\omega^2$.
- B. A. Nikolaus, T. Hoch and D. G. Madland, Phys. Rev. C 46 (1992) 1757 — relativistic point-coupling (contact) models, with four-fermion terms of the form $-\frac{\alpha_S}{2}(\bar\psi\psi)^2 - \frac{\alpha_V}{2}(\bar\psi\gamma_\mu\psi)(\bar\psi\gamma^\mu\psi)$, the 3+1-dimensional analogue of the g_S and g_V terms here.
- N. M. Hugenholtz and L. Van Hove, Physica 24 (1958) 363 — at saturation (zero pressure $P = \rho\mu - \varepsilon$) the Fermi energy equals the energy per particle, $\varepsilon_F = \mu = E/A$.
- G. C. Wick, Phys. Rev. 80 (1950) 268 — expectation values in a Slater determinant factorize into contractions: $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle = \varrho_{ba}\varrho_{dc} - \varrho_{da}\varrho_{bc}$, $\varrho_{xy} = \langle c_y^\dagger c_x\rangle$.
- M. Fierz, Z. Phys. 104 (1937) 553 — rearrangement of four-fermion products from the completeness of the Clifford basis, here $\sum_K (\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\,\delta_{ad}\delta_{cb}$.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345; S. P. Klevansky, Rev. Mod. Phys. 64 (1992) 649 — chirally symmetric contact coupling $G[(\bar\psi\psi)^2 + (\bar\psi i\gamma_5\psi)^2]$ with a momentum cutoff; gap equation $M = m - 2G\langle\bar\psi\psi\rangle$; for $m = 0$ a solution $M \neq 0$ exists above a critical coupling (here $G = -g/2$).
- F. A. Berezin, *The Method of Second Quantization*, Academic Press (1966) — Grassmann algebra, $\theta_k\theta_l = -\theta_l\theta_k$, and the left derivative.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989) — the classification of real Clifford algebras. In the convention used here ($Cl(p,q)$ has p generators squaring to $+1$ and q to -1) it gives $Cl(1,7) \cong \mathbb{R}(16)$ and $Cl(7,1) \cong \mathbb{H}(8)$; Lawson and Michelsohn define $v^2 = -q(v)$, so their $Cl_{7,1}$ is the $Cl(1,7)$ of this record.

3. Decisions and derivations

Each subsection states a decision or a derived result, the equations behind it, and the checks of 7+1.nb that confirm it (quoted by their names in section 4.2).

3.1 The real Dirac matrices and the sign of the metric

Decision. The Dirac matrices are the real generators Γ_A of section 2.1, with

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \eta = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1): \qquad \Gamma_0^2 = 1_{16}, \quad \Gamma_i^2 = -1_{16}.$$

Reason. The task requires real matrices, and only this overall sign allows them. With the opposite sign $\eta' = -\eta$ the relations $R_AR_B + R_BR_A = 2\eta'_{AB}1_{16}$ define $Cl(7,1) \cong \mathbb{H}(8)$. If real 16×16 matrices R_A satisfied them, then, because $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has only one irreducible module, $R_A = T(i\Gamma_A)T^{-1}$ for an invertible complex T ($i\Gamma_A$ satisfy the same relations). $R_A = \overline{R_A}$ gives $Y\overline{(i\Gamma_A)} = (i\Gamma_A)Y$ with $Y = T^{-1}\overline{T}$, $Y\overline{Y} = 1_{16}$; since $\overline{(i\Gamma_A)} = -i\Gamma_A$ this is $Y\Gamma_A = -\Gamma_A Y$, so $Y\omega$ commutes with all Γ_A and $Y = c\,\omega$ (Schur), whence $Y\overline{Y} = |c|^2\omega^2 = -|c|^2 1_{16} \neq 1_{16}$. Renaming, reordering or re-signing the indices does not change the signature. The first stage proved this in 7+1.nb.

The Dirac matrices with upper index, α, β, ω (checks 1–5 of section 7.1 of 7+1.nb):

$$\Gamma^A = \eta^{AA}\Gamma_A, \qquad (\Gamma^A)^\dagger = \Gamma^0\Gamma^A\Gamma^0, \qquad \beta = \Gamma^0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix},$$

$$\alpha_i = \Gamma^0\Gamma^i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix}, \qquad \alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}1_{16}, \qquad \alpha_i\beta + \beta\alpha_i = 0, \qquad \beta^2 = 1_{16},$$

$$\omega = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \omega^2 = -1_{16}, \qquad \omega\Gamma_A = -\Gamma_A\omega.$$

3.2 The Dirac equation, the Klein–Gordon equation and the Hamiltonian

Decision. $(i\Gamma^A\partial_A - m)\psi = 0$. The factor i is unavoidable: with real Γ_A, $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \Box$, so

$$\big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\Box + m^2\big)1_{16},$$

which is the Klein–Gordon operator with the correct sign for a massive particle; without the i the mass term would have the wrong sign ($(\Gamma^A\partial_A)^2 - m^2 = \Box - m^2$). Multiplying by Γ^0:

$$\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - h\big)\psi, \qquad h = -i\alpha_i\partial_i + \beta m, \qquad h^\dagger = h,$$

because α_i, β are real symmetric and $-i\partial_i$ is Hermitian. Conservation of $j^A = \bar\psi\Gamma^A\psi$, $\bar\psi = \psi^\dagger\Gamma^0$, follows from the adjoint equation $-i\partial_A\bar\psi\Gamma^A - m\bar\psi = 0$: $\partial_A j^A = (im\bar\psi)\psi + \bar\psi(-im\psi) = 0$. 7+1.nb checks these identities for 16 arbitrary functions $\psi_a(x_0,\dots,x_7)$.

3.3 Real plane-wave spinors

For $e^{-i(Ex_0 - \vec p\cdot\vec x)}$, h becomes the real symmetric matrix

$$h(\vec p) = p^i\alpha_i + m\beta = \begin{pmatrix} 0 & M \\ M^{\mathsf T} & 0 \end{pmatrix}, \qquad M = m1_8 + p^i\tau_i^O,$$

$$MM^{\mathsf T} = m^2 1_8 - (p^i\tau_i^O)^2 = E^2 1_8, \qquad E = \sqrt{m^2 + \vec p^{\,2}},$$

using $(p^i\tau_i^O)^{\mathsf T} = -p^i\tau_i^O$ and $(p^i\tau_i^O)^2 = -\vec p^{\,2}1_8$. **Derivation of the spinors.** With $u_s = \frac{1}{\sqrt E}(E\,e_s;\ M^{\mathsf T}e_s)$ and $v_s = \frac{1}{\sqrt E}(E\,e_s;\ -Me_s)$, and $h(-\vec p) = \left(\begin{smallmatrix} 0 & M^{\mathsf T} \\ M & 0 \end{smallmatrix}\right)$:

$$h(\vec p)\,(E\,e_s;\ M^{\mathsf T}e_s) = (MM^{\mathsf T}e_s;\ E\,M^{\mathsf T}e_s) = E\,(E\,e_s;\ M^{\mathsf T}e_s),$$

$$h(-\vec p)\,(E\,e_s;\ -Me_s) = (-M^{\mathsf T}Me_s;\ E\,Me_s) = -E\,(E\,e_s;\ -Me_s),$$

$$u_s^{\mathsf T}u_r = \frac{E^2\delta_{sr} + e_s^{\mathsf T}MM^{\mathsf T}e_r}{E} = 2E\,\delta_{sr}, \qquad v_s^{\mathsf T}v_r = 2E\,\delta_{sr}.$$

Decision (normalization). $u^{\mathsf T}u = 2E$, the usual covariant normalization, so that $\sum_s u_s\bar u_s = \not p + m$ and $\sum_s v_s\bar v_s = \not p - m$ with $\not p = E\Gamma_0 + p^i\Gamma_i$. The example $\vec p = (2,2,4,0,0,0,0)$, $m = 1$, $E = 5$ was chosen because E is an integer, so $\sqrt5\,u_s$ has integer entries. The real orthogonal matrix $U = \frac{1}{\sqrt{2E}}\big(u_1(\vec p),\dots,u_8(\vec p),v_1(-\vec p),\dots,v_8(-\vec p)\big)$ satisfies $U^{\mathsf T}h(\vec p)U = \mathrm{diag}(E1_8, -E1_8)$, equivalently the completeness relation

$$\sum_{s=1}^{8} u_s(\vec p)\,u_s(\vec p)^{\mathsf T} + \sum_{s=1}^{8} v_s(-\vec p)\,v_s(-\vec p)^{\mathsf T} = 2E\,1_{16}.$$

3.4 Grassmann algebra, quantization and the canonical anticommutators

Decision. The Grassmann algebra is implemented with ordered monomials `grass[{k1, …, kr}]`, the product $\theta_K\theta_{K'} = \mathrm{sign}(\pi)\theta_{\mathrm{sort}(K\cup K')}$ (zero if $K\cap K' \neq \emptyset$) and the left derivative $\partial^L(\theta_{k_1}\cdots\theta_{k_r})/\partial\theta_{k_j} = (-1)^{j-1}\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r}$, as in Berezin. The current-conservation check uses $\psi = \sum_{k=1}^{3}\theta_k U_k$, $\bar\psi = \sum_k \theta_{3+k}U_k^\dagger\Gamma^0$ with

$$U_1 = u_1(\vec p_1)e^{-i(5x_0 - \vec p_1\cdot\vec x)}, \qquad U_2 = v_3(\vec p_3)e^{+i(3x_0 - \vec p_3\cdot\vec x)}, \qquad U_3 = u_5(\vec 0)e^{-ix_0},$$

$$\vec p_1 = (2,2,4,0,0,0,0), \qquad \vec p_3 = (0,0,0,2,2,0,0).$$

The mode expansion and the anticommutators of the modes

$$\psi(x_0,\vec x) = \sum_{\vec p}\sum_{s=1}^{8}\frac{1}{\sqrt{2EV}}\Big[b_s(\vec p)u_s(\vec p)e^{-i(Ex_0 - \vec p\cdot\vec x)} + d_s^\dagger(\vec p)v_s(\vec p)e^{+i(Ex_0 - \vec p\cdot\vec x)}\Big],$$

$$\{b_s(\vec p), b_r^\dagger(\vec q)\} = \{d_s(\vec p), d_r^\dagger(\vec q)\} = \delta_{sr}\delta_{\vec p\vec q} \quad \text{(others 0)}$$

give, with the completeness relation of section 3.3 (the v term contributes at $-\vec p$, which the symmetric sum over $\vec p$ relabels),

$$\begin{aligned}\{\psi_a(x_0,\vec x), \psi_b^\dagger(x_0,\vec y)\} &= \frac{1}{V}\sum_{\vec p}\frac{e^{i\vec p\cdot(\vec x-\vec y)}}{2E}\Big[\sum_s u_s u_s^{\mathsf T} + \sum_s v_s(-\vec p)v_s(-\vec p)^{\mathsf T}\Big]_{ab}\\ &= \delta_{ab}\frac{1}{V}\sum_{\vec p} e^{i\vec p\cdot(\vec x-\vec y)} = \delta_{ab}\,\delta^7(\vec x-\vec y).\end{aligned}$$

Decision (test set). The exact check uses five momenta closed under $\vec p \to -\vec p$, $\pm(2,2,4,0,0,0,0)$, $\pm(0,0,0,2,2,0,0)$, $\vec 0$ ($m = 1$, $E = 5,5,3,3,1$, all integers), with symbolic $\vec x$, $\vec y$ and box volume V. The normal-ordered Hamiltonian and charge follow from $U^{\mathsf T}hU = \mathrm{diag}(E,-E)$: $H = \sum E(b^\dagger b + d^\dagger d)$, $Q = \sum(b^\dagger b - d^\dagger d)$.

3.5 Charge conjugation

Decision. $\psi^c = \omega\psi^*$. Because the Γ_A are real, conjugating the Dirac equation gives $(-i\Gamma^A\partial_A - m)\psi^* = 0$, and ω is real and anticommutes with every Γ_A:

$$\big(i\Gamma^A\partial_A - m\big)\omega\psi^* = \omega\big(-i\Gamma^A\partial_A - m\big)\psi^* = 0, \qquad (\psi^c)^c = \omega\omega\psi = -\psi, \qquad (\not p + m)\,\omega\,u_s(\vec p) = 0.$$

Consequently $\psi^c = \psi$ has no non-zero solution: a massive real (Majorana) field needs 32 real components in 7+1 dimensions.

3.6 Lorentz covariance and the boost example

The spinor boost $\mathcal B(\beta_{\mathrm b}) = e^{\beta_{\mathrm b}\Sigma_{01}}$ and its Lorentz matrix L (section 2.1) satisfy $\mathcal B\Gamma_K\mathcal B^{-1} = \sum_B L_{BK}\Gamma_B$ and $L^{\mathsf T}\eta L = \eta$, with

$$L = \begin{pmatrix}\cosh\beta_{\mathrm b} & -\sinh\beta_{\mathrm b} & 0\\ -\sinh\beta_{\mathrm b} & \cosh\beta_{\mathrm b} & 0\\ 0 & 0 & 1_6\end{pmatrix}.$$

Decision (example). $\beta_{\mathrm b} = 2\log 2$, because then every number is rational: $\cosh\frac{\beta_{\mathrm b}}{2} = \frac54$, $\sinh\frac{\beta_{\mathrm b}}{2} = \frac34$, $\cosh\beta_{\mathrm b} = \frac{17}{8}$, $\sinh\beta_{\mathrm b} = \frac{15}{8}$. For $p = (5,2,2,4,0,0,0,0)$, $m = 1$:

$$p' = Lp = \Big(\tfrac{17}{8}\cdot 5 - \tfrac{15}{8}\cdot 2,\ -\tfrac{15}{8}\cdot 5 + \tfrac{17}{8}\cdot 2,\ 2,4,0,0,0,0\Big) = \Big(\tfrac{55}{8}, -\tfrac{41}{8}, 2,4,0,0,0,0\Big), \qquad p'\cdot p' = \tfrac{3025-1681}{64} - 20 = 1,$$

and $(\not p' - m)\,\mathcal B\,u_s(\vec p) = 0$ for $s = 1,\dots,8$, with $\mathcal B = \frac54\,1_{16} + \frac34\,\Gamma_0\Gamma_1$.

3.7 Chirality and the two coupled 8-component equations

Result. No real chirality matrix exists: the real 16×16 matrices Z with $Z\Gamma_A = -\Gamma_A Z$ ($A = 0,\dots,7$) form the space $\mathbb R\,\omega$ (7+1.nb solves the 256-unknown linear system, rank 255), and $(c\,\omega)^2 = -c^2 1_{16}$ cannot be 1_{16}. **Decision.** Chirality is the Hermitian matrix $\chi = i\omega$, unique up to sign; the sign is chosen so that $P_L = \frac12(1_{16} - \chi)$ projects on $\chi = -1$:

$$\chi = \begin{pmatrix}0 & -i1_8\\ i1_8 & 0\end{pmatrix}, \qquad \chi^2 = 1_{16}, \qquad \chi\Gamma_A = -\Gamma_A\chi, \qquad \chi\Sigma_{AB} = \Sigma_{AB}\chi,$$

$$E_L = \frac{1}{\sqrt2}\begin{pmatrix}1_8\\ -i1_8\end{pmatrix}, \quad E_R = \frac{1}{\sqrt2}\begin{pmatrix}-i1_8\\ 1_8\end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R, \quad E_L E_L^\dagger = P_L, \quad E_R E_R^\dagger = P_R.$$

Decision (chiral components). $\phi_L = E_L^\dagger\psi = (\psi_\uparrow + i\psi_\downarrow)/\sqrt2$ and $\phi_R = E_R^\dagger\psi = (i\psi_\uparrow + \psi_\downarrow)/\sqrt2$. These isometries were chosen because they make the chiral blocks of the Dirac matrices as simple as possible:

$$E_L^\dagger\beta E_R = E_R^\dagger\beta E_L = 1_8, \qquad E_L^\dagger\beta E_L = E_R^\dagger\beta E_R = 0,$$

$$E_L^\dagger \alpha_i E_L = -\sigma_i, \qquad E_R^\dagger \alpha_i E_R = \sigma_i, \qquad E_L^\dagger \alpha_i E_R = 0, \qquad \sigma_i = i\tau_i^O.$$

Derivation of the chiral equations. Insert $\psi = E_L\phi_L + E_R\phi_R$ into $\Gamma^0(i\Gamma^A\partial_A - m)\psi = (i\partial_0 + i\alpha_i\partial_i - \beta m)\psi$ and project with $E_L^\dagger$, $E_R^\dagger$; since $i\,(-\sigma_i)\partial_i = \tau_i^O\partial_i$:

$$E_L^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 + \tau_i^O\partial_i\big)\phi_L - m\,\phi_R,$$
$$E_R^\dagger\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - \tau_i^O\partial_i\big)\phi_R - m\,\phi_L.$$

So the Dirac equation is equivalent to $(i\partial_0 + \tau_i^O\partial_i)\phi_L = m\phi_R$ and $(i\partial_0 - \tau_i^O\partial_i)\phi_R = m\phi_L$, coupled exactly when $m \neq 0$, with only real 8×8 matrices besides the i of $i\partial_0$. Each obeys the Klein–Gordon equation because $(i\partial_0 - \tau_i^O\partial_i)(i\partial_0 + \tau_j^O\partial_j) = -\Box 1_8$. For a plane wave the equations read $(E + \sigma\cdot\vec{p})\phi_L = m\phi_R$, $(E - \sigma\cdot\vec{p})\phi_R = m\phi_L$, and $W = (E_L, E_R)$ gives $W^\dagger h(\vec{p})W = \left(\begin{smallmatrix} -\sigma\cdot\vec{p} & m1_8 \\ m1_8 & \sigma\cdot\vec{p}\end{smallmatrix}\right)$.

3.8 The self-coupled field

Decision (interaction). The field is coupled to itself by the four Lorentz-invariant contact (zero-range) interactions built from bilinears without derivatives that are used in relativistic point-coupling models: scalar, pseudoscalar, vector and axial vector,

$$\begin{aligned}\mathcal{L} = {}& \bar\psi\big(i\Gamma^A\partial_A - m\big)\psi - \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi \\ & - \frac{g_S}{2}(\bar\psi\psi)^2 - \frac{g_P}{2}(\bar\psi\, i\chi\psi)^2 - \frac{g_V}{2}(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi) - \frac{g_A}{2}(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi),\end{aligned}$$

with real external sources $\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$ that define the densities. The tensor channel $(\bar\psi\Sigma_{AB}\psi)^2$ and the channels of higher rank are not included; they could be added in the same way, and by the Fierz identities of section 3.11 their exchange terms are again combinations of direct terms. The four densities are real because $\Gamma^0 O$ is Hermitian for $O = 1_{16},\ i\chi,\ \Gamma^A,\ \Gamma^A\chi$. **Units.** $\hbar = c = 1$; ψ has mass dimension 7/2 and every g mass dimension −6, so the theory is an effective one and needs either the no-sea approximation (section 3.10) or a momentum cutoff (section 3.14). **Signs.** The interaction has no time derivatives, so it enters the Hamiltonian density with the opposite sign:

$$\begin{aligned}\mathcal{H} = {}& :\psi^\dagger(-i\alpha_i\partial_i + \beta m)\psi: + :\bar\psi\big(\mathsf{s} + i\chi\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\psi: \\ & + \frac{g_S}{2} :\rho_S^2: + \frac{g_P}{2} :\rho_P^2: + \frac{g_V}{2} :j^A j_A: + \frac{g_A}{2} :j_5^A j_{5A}:,\end{aligned}$$

with $\rho_S = \bar\psi\psi$, $\rho_P = \bar\psi i\chi\psi$, $j^A = \bar\psi\Gamma^A\psi$, $j_5^A = \bar\psi\Gamma^A\chi\psi$; $g_S < 0$ attracts.

3.9 The nonlinear field equation

The Euler–Lagrange equation is taken with the Grassmann LEFT derivative with respect to $\bar\psi_a$; the bilinears are Grassmann-even, so

$$\begin{aligned}&\big(i\Gamma^A\partial_A - m - \mathsf{s} - i\chi\mathsf{p} - \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\psi \\ &\quad = g_S(\bar\psi\psi)\psi + g_P(\bar\psi i\chi\psi)\, i\chi\psi + g_V(\bar\psi\Gamma_A\psi)\Gamma^A\psi + g_A(\bar\psi\Gamma_A\chi\psi)\Gamma^A\chi\psi.\end{aligned}$$

Decision (check). 7+1.nb verifies $\partial^L\mathcal{L}_{\rm int}/\partial\bar\psi_a$ for all 16 components in a Grassmann algebra with the 32 generators $\psi_a = \theta_a$, $\bar\psi_a = \theta_{16+a}$, with symbolic couplings.

3.10 Hohenberg–Kohn theorem and Kohn–Sham equations

With $n = (\rho_S, \rho_P, j^A, j_5^A)$ (18 densities), $w = (\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$ and $H[w] = H_0 + H_{\rm int} + \int w\cdot n\, d^7x$ in the sector of fixed charge $Q = N$, two sources $w \neq w'$ with different non-degenerate ground states Ψ, Ψ' and the same densities would give

$$E < \langle\Psi'|H[w]|\Psi'\rangle = E' + \int (w - w')\cdot n, \qquad E' < E + \int (w' - w)\cdot n \qquad \Rightarrow \qquad E + E' < E + E',$$

a contradiction; so $E[w]=\min_n\big(F[n]+\int w\cdot n\big)$ with $F[n]=\min_{\Psi\to n}\langle\Psi|H_0+H_{\rm int}|\Psi\rangle$. Writing $F=T_s+E_H+E_{xc}$ with

$$T_s=\sum_{k=1}^{N}\int\varphi_k^\dagger(-i\alpha_i\partial_i+\beta m)\varphi_k,\qquad E_H=\int\Big[\frac{g_S}{2}\rho_S^2+\frac{g_P}{2}\rho_P^2+\frac{g_V}{2}j^Aj_A+\frac{g_A}{2}j_5^Aj_{5A}\Big],$$

and varying $E=T_s+\int w\cdot n+E_H+E_{xc}$ with respect to $\varphi_k^\dagger$ under $\int\varphi_k^\dagger\varphi_l=\delta_{kl}$ (multipliers ε_k) gives $h_{\rm KS}\varphi_k=\varepsilon_k\varphi_k$ with

$$h_{\rm KS}=\alpha_i(-i\partial_i+\mathcal{V}_i)+\beta(m+\mathcal{S})+\mathcal{V}_0+\beta i\chi\mathcal{P}+\chi\mathcal{A}_0+\alpha_i\chi\mathcal{A}_i,$$
$$\mathcal{S}=\mathsf{s}+g_S\rho_S+\frac{\delta E_{xc}}{\delta\rho_S},\quad \mathcal{P}=\mathsf{p}+g_P\rho_P+\frac{\delta E_{xc}}{\delta\rho_P},\quad \mathcal{V}_A=\mathsf{v}_A+g_Vj_A+\frac{\delta E_{xc}}{\delta j^A},\quad \mathcal{A}_A=\mathsf{a}_A+g_Aj_{5A}+\frac{\delta E_{xc}}{\delta j_5^A}.$$

The variation uses $\delta\rho_S/\delta\varphi^\dagger=\beta\varphi$, $\delta\rho_P/\delta\varphi^\dagger=\beta i\chi\varphi$, $\delta j^A/\delta\varphi^\dagger=\Gamma^0\Gamma^A\varphi$, $\delta j_5^A/\delta\varphi^\dagger=\Gamma^0\Gamma^A\chi\varphi$, so the Hartree part is $v_H=\beta g_S\rho_S+\beta i\chi g_P\rho_P+\Gamma^0\Gamma^Ag_Vj_A+\Gamma^0\Gamma^A\chi g_Aj_{5A}$ (checked as Wirtinger derivatives, $\partial e_H/\partial\varphi_a^*=(v_H\varphi)_a$, with 32 independent variables). **Decision (no sea).** The occupied orbitals are the N lowest positive-energy states; the total energy is

$$E=\sum_k\varepsilon_k-\int\big[(\mathcal{S}-\mathsf{s})\rho_S+(\mathcal{P}-\mathsf{p})\rho_P+(\mathcal{V}_A-\mathsf{v}_A)j^A+(\mathcal{A}_A-\mathsf{a}_A)j_5^A\big]+E_H+E_{xc}.$$

3.11 Exchange (Wick) and the Fierz identities

For a Slater determinant of N orbitals and contact vertices, Wick's theorem with $\varrho_{xy}=\sum_k\varphi_{k,x}\varphi_{k,y}^*$ and $R=\varrho\Gamma^0$ gives

$$\langle:(\bar\psi O_1\psi)(\bar\psi O_2\psi):\rangle=\operatorname{tr}(O_1R)\operatorname{tr}(O_2R)-\operatorname{tr}(O_1RO_2R),\qquad R=\sum_k\varphi_k\bar\varphi_k,$$
$$E_x=-\frac{g_S}{2}\operatorname{tr}(RR)-\frac{g_P}{2}\operatorname{tr}(i\chi R\,i\chi R)-\frac{g_V}{2}\operatorname{tr}(\Gamma^AR\Gamma_AR)-\frac{g_A}{2}\operatorname{tr}(\Gamma^A\chi R\Gamma_A\chi R),$$
$$\Sigma_x=-g_SR-g_P(i\chi)R(i\chi)-g_V\Gamma_AR\Gamma^A-g_A\Gamma_A\chi R\Gamma^A\chi,\qquad \delta E_x=\operatorname{tr}(\Sigma_x\delta R).$$

Decision (check). 7+1.nb builds the 560-dimensional Fock space of 3 particles in 16 modes, with $c_a^\dagger c_b$ as explicit sparse matrices and $:B(M_1)B(M_2):=B(M_1)B(M_2)-B(M_1M_2)$, $B(M)=\sum M_{ab}c_a^\dagger c_b$, $M_i=\Gamma^0O_i$, and checks the formula for all 18 vertex pairs with 3 random orthonormal orbitals (`SeedRandom[71]`, 40 digits). The Fierz completeness $\sum_K(\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd}=16\delta_{ad}\delta_{cb}$ gives the rearrangement

$$\operatorname{tr}(O_1RO_2R)=\frac{1}{16}\sum_K\operatorname{tr}(\Gamma_K^{-1}R)\operatorname{tr}(O_1\Gamma_KO_2R),$$

checked with random complex integer matrices R, δR (`SeedRandom[1771]`). **Uniform matter.** With $R=(\rho_S1_{16}+\rho\Gamma_0)/16$, using $\chi R\chi=(\rho_S-\rho\Gamma_0)/16$, $\Gamma^A\Gamma_A=8$ and $\Gamma^A\Gamma_0\Gamma_A=\Gamma_0+7\cdot(-\Gamma_0)=-6\Gamma_0$:

$$\operatorname{tr}(RR)=\frac{\rho_S^2+\rho^2}{16},\qquad \operatorname{tr}(i\chi R\,i\chi R)=-\frac{\rho_S^2-\rho^2}{16},$$
$$\operatorname{tr}(\Gamma^AR\Gamma_AR)=\frac{8\rho_S^2-6\rho^2}{16},\qquad \operatorname{tr}(\Gamma^A\chi R\Gamma_A\chi R)=-\frac{8\rho_S^2+6\rho^2}{16},$$
$$E_x=-\frac{g_S}{32}(\rho_S^2+\rho^2)+\frac{g_P}{32}(\rho_S^2-\rho^2)-\frac{g_V}{32}(8\rho_S^2-6\rho^2)+\frac{g_A}{32}(8\rho_S^2+6\rho^2),$$
$$G_S=g_S+\frac{-g_S+g_P-8g_V+8g_A}{16},\qquad G_V=g_V+\frac{-g_S-g_P+6g_V+6g_A}{16},$$
$$\frac{g_S}{2}\rho_S^2+\frac{g_V}{2}\rho^2+E_x=\frac{G_S}{2}\rho_S^2+\frac{G_V}{2}\rho^2$$

(the pseudoscalar and axial Hartree terms vanish for uniform matter, where $\rho_P=j_5^A=0$). For $g_S=-10^5$, $g_V=10^4$, $g_P=g_A=0$: $G_S=-10^5+\frac{10^5-8\cdot10^4}{16}=-98750$ and $G_V=10^4+\frac{10^5+6\cdot10^4}{16}=20000$.

3.12 The chiral Kohn–Sham theory

Decision. Use the variables $\kappa = \sum_k \varphi^\dagger_{kL}\varphi_{kR}$, $j^A_L = \sum_k \varphi^\dagger_{kL}\bar\sigma^A\varphi_{kL}$, $j^A_R = \sum_k \varphi^\dagger_{kR}\sigma^A\varphi_{kR}$ ($\sigma^A = (1_8, \sigma_i)$, $\bar\sigma^A = (1_8, -\sigma_i)$), the complex source $\varpi = \mathsf{s} + i\mathsf{p}$ and the complex mass $\mathcal{M} = m + \mathcal{S} + i\mathcal{P}$. From the chiral blocks of section 3.7:

$$\rho_S = \kappa + \kappa^*, \quad \rho_P = i(\kappa - \kappa^*), \quad j^A = j^A_L + j^A_R, \quad j^A_5 = j^A_R - j^A_L, \quad \mathsf{s}\rho_S + \mathsf{p}\rho_P = \varpi\kappa + \varpi^*\kappa^*,$$
$$e_H = \frac{g_S - g_P}{2}(\kappa^2 + \kappa^{*2}) + (g_S + g_P)\kappa\kappa^* + \frac{g_V + g_A}{2}(j^A_L j_{LA} + j^A_R j_{RA}) + (g_V - g_A) j^A_L j_{RA},$$
$$W^\dagger h_{\mathrm{KS}}(\vec p) W = \begin{pmatrix} \mathcal{V}^L_0 - \sigma_i(p^i + \mathcal{V}^L_i) & \mathcal{M}1_8 \\ \mathcal{M}^* 1_8 & \mathcal{V}^R_0 + \sigma_i(p^i + \mathcal{V}^R_i) \end{pmatrix}, \qquad \mathcal{V}^{L,R}_A = \mathcal{V}_A \mp \mathcal{A}_A,$$
$$\mathcal{M} - m = \varpi + \frac{\partial(e_H + e_{xc})}{\partial\kappa}, \qquad \text{Hartree: } \mathcal{M} - m - \varpi = g_S(\kappa + \kappa^*) - g_P(\kappa - \kappa^*) = g_S\rho_S + i g_P \rho_P.$$

Notation decision. The complex source is called ϖ, not μ: μ is the chemical potential of section 3.13. **Chiral symmetry.** $e^{i\theta\chi} = \cos\theta\, 1_{16} - \sin\theta\,\omega$ is real and multiplies κ by $e^{2i\theta}$; E_H changes by $\frac{g_S - g_P}{2}[(e^{4i\theta} - 1)\kappa^2 + (e^{-4i\theta} - 1)\kappa^{*2}]$, so the interaction is chirally symmetric exactly for $g_P = g_S$. The check uses $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$ (rational).

3.13 Uniform matter: formulas, parameters and numerical precision

Static uniform matter at rest, all 8 spin states filled to k_F, no sea, no sources: $\mathcal{S} = m^* - m$, $\mathcal{V}_0$ constant, $\mathcal{P} = \mathcal{A}_A = \mathcal{V}_i = 0$. The area of the unit sphere $S^6 \subset \mathbb{R}^7$ is $|S^6| = 2\pi^{7/2}/\Gamma_{\mathrm{E}}(\frac{7}{2}) = \frac{16\pi^3}{15}$ (Γ_{E} is Euler's Gamma function, $\Gamma_{\mathrm{E}}(\frac{7}{2}) = \frac{15}{8}\sqrt{\pi}$), so

$$8\int \frac{d^7k}{(2\pi)^7} f(|\vec k|) = c_7 \int_0^\infty k^6 f(k)\, dk, \qquad c_7 = \frac{8|S^6|}{(2\pi)^7} = \frac{1}{15\pi^4},$$
$$\rho = c_7 \frac{k_F^7}{7} = \frac{k_F^7}{105\pi^4}, \qquad \rho_S = c_7 \int_0^{k_F} \frac{k^6 m^*}{\sqrt{k^2 + m^{*2}}} dk, \qquad \varepsilon_{\mathrm{kin}} = c_7 \int_0^{k_F} k^6 \sqrt{k^2 + m^{*2}}\, dk,$$
$$m^* = m + g_S\rho_S(m^*), \qquad \mathcal{V}_0 = g_V\rho, \qquad \varepsilon_F = \mathcal{V}_0 + \sqrt{k_F^2 + m^{*2}},$$
$$\varepsilon = \varepsilon_{\mathrm{kin}} - \frac{(m^* - m)^2}{2g_S} + \frac{g_V}{2}\rho^2.$$

The integrals have the closed forms (obtained with `Integrate` in 7+1.nb, checked by differentiation with sympy), with $\operatorname{artanh} y = \frac{1}{2}\log\frac{1+y}{1-y}$:

$$\int_0^{k_F} \frac{k^6 m^*}{\sqrt{k^2 + m^{*2}}} dk = \frac{m^*}{48}\Big[k_F\sqrt{k_F^2 + m^{*2}}\big(8k_F^4 - 10k_F^2 m^{*2} + 15m^{*4}\big) - 15m^{*6}\operatorname{artanh}\frac{k_F}{\sqrt{k_F^2 + m^{*2}}}\Big],$$
$$\int_0^{k_F} k^6\sqrt{k^2 + m^{*2}}\, dk = \frac{1}{384}\Big[k_F\sqrt{k_F^2 + m^{*2}}\big(48k_F^6 + 8k_F^4 m^{*2} - 10k_F^2 m^{*4} + 15m^{*6}\big) - 15m^{*8}\operatorname{artanh}\frac{k_F}{\sqrt{k_F^2 + m^{*2}}}\Big].$$

Hugenholtz–Van Hove. $\partial\varepsilon/\partial m^* = \rho_S - (m^* - m)/g_S = 0$ is the gap equation (with $d\varepsilon_{\mathrm{kin}}/dm^* = \rho_S$), so $d\varepsilon/dk_F = c_7 k_F^6(\sqrt{k_F^2 + m^{*2}} + g_V\rho)$ and, with $d\rho/dk_F = c_7 k_F^6$, the chemical potential is $\mu = d\varepsilon/d\rho = \varepsilon_F$; then $\frac{d}{dk_F}\frac{\varepsilon}{\rho} = \frac{c_7 k_F^6}{\rho}\big(\mu - \frac{\varepsilon}{\rho}\big)$ vanishes at the minimum of $E/A = \varepsilon/\rho$, where $E/A = \varepsilon_F$.

Decisions (parameters). Units $m = 1$. $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$ (couplings of mass dimension -6): with these values the energy per particle has a minimum below m at $k_F = 0.7566\ldots$, so the example shows binding and saturation; the Hartree–Fock comparison is made at $k_F = \frac{3}{4}$. **Decisions (precision).** The gap equation is solved by Brent's method with 50-digit working precision (`wp = 50`, accuracy and precision goals 45), the input k_F padded to 60 digits; the saturation condition $\varepsilon/\rho = \varepsilon_F$ is solved by Brent's method on $[0.7, 0.8]$ with 40-digit working precision and goals 32. The checks require $|E/A - \varepsilon_F| < 10^{-25}$ at saturation and compare μ from a symmetric difference quotient (step $h = 10^{-12}$ in k_F) with ε_F to 10^{-15}. These settings

were adopted after development runs in which `FindRoot` issued the message `FindRoot::precw` (input precision below the working precision) and its default goals (half the working precision) left residuals larger than the tolerances of the checks.

3.14 The chiral vacuum with the Dirac sea

Decision. To show spontaneous chiral symmetry breaking, keep the Dirac sea with a momentum cutoff $|\vec{p}| < \Lambda$, take $g_P = g_S = g$, $\varpi = 0$, $\mathcal{V} = \mathcal{A} = 0$ and a uniform complex mass $\mathcal{M}$:

$$h_\chi(\vec{p}) = \begin{pmatrix} -\sigma\cdot\vec{p} & \mathcal{M}1_8 \\ \mathcal{M}^* 1_8 & \sigma\cdot\vec{p} \end{pmatrix}, \qquad h_\chi^2 = (\vec{p}^{\,2} + |\mathcal{M}|^2)1_{16}, \qquad \Pi_- = \frac{1}{2}\Big(1_{16} - \frac{h_\chi}{E}\Big), \quad E = \sqrt{\vec{p}^{\,2} + |\mathcal{M}|^2},$$

$$\kappa_{\vec{p}} = \operatorname{tr}\Pi_-^{RL} = -\frac{4\mathcal{M}^*}{E}, \qquad \frac{|S^6|}{(2\pi)^7} = \frac{1}{120\pi^4}, \qquad \kappa = -4\mathcal{M}^* J(|\mathcal{M}|),$$

$$J(M) = \frac{1}{120\pi^4}\int_0^\Lambda \frac{p^6 dp}{\sqrt{p^2+M^2}}, \qquad K(M) = \frac{1}{120\pi^4}\int_0^\Lambda p^6\sqrt{p^2+M^2}\,dp,$$

$$\mathcal{M} = m + 2g\kappa^* = m - 8gJ(|\mathcal{M}|)\mathcal{M}, \qquad \mathcal{E}(\mathcal{M}) = -8K + 8J\,\mathrm{Re}\big[(\mathcal{M}-m)\mathcal{M}^*\big] + 32g|\mathcal{M}|^2J^2.$$

Here Π_-^{RL} is the lower-left 8×8 block of Π_-, equal to $-\mathcal{M}^*1_8/(2E)$, and $\mathcal{E}$ is the sum of the sea eigenvalues $-8K$ minus the double counting $2\mathrm{Re}[(\mathcal{M}-m)\kappa]$ plus $e_H = 2g|\kappa|^2$. For $m = 0$ a solution $\mathcal{M} \neq 0$ needs $1 + 8gJ(|\mathcal{M}|) = 0$, possible exactly when

$$g < g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda^6}, \qquad J(0) = \frac{\Lambda^6}{720\pi^4}, \qquad K(0) = \frac{\Lambda^8}{960\pi^4}.$$

Decisions (parameters). $\Lambda = 1$; $g = \frac{6}{5}g_c$ (20% beyond the critical coupling, so that $|\mathcal{M}|$ is clearly non-zero); a small explicit mass $m = \frac{1}{100}$ to show that it selects the real positive $\mathcal{M}$, $\mathcal{E}(\mathcal{M}e^{i\phi}) - \mathcal{E}(\mathcal{M}) = 8Jm|\mathcal{M}|(1-\cos\phi) > 0$; the diagonalization check at $\vec{p} = (\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{4}, 0, -\frac{3}{20}, \frac{1}{20})$ with $\mathcal{M} = |\mathcal{M}|e^{7i/10}$ (a generic momentum and phase). The gap equation is solved by Brent's method with 40-digit working precision and goals 32.

3.15 Notation decisions

- The coordinates $x_0, \ldots, x_7$ and the metric η of 7+1.nb are kept. Kohn–Sham potentials are calligraphic ($\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A$), sources sans-serif ($\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$).
- The Dirac matrix $\beta = \Gamma^0$ keeps its name; the rapidity of a boost is β_b (Mathematica `β` in `Sβ`), the spinor boost is $\mathcal{B}(\beta_\mathrm{b})$ (Mathematica `Sβ`) and its Lorentz matrix L (Mathematica `Λboost`), so that $\mathcal{S}$ means only the scalar potential and Λ only the cutoff.
- The complex scalar source is $\varpi = \mathsf{s} + i\mathsf{p}$; μ is only the chemical potential.
- The matrices of the two proofs are Y (no real realization for η') and Z (no real chirality); J and K are only the sea integrals; Euler's Gamma function is Γ_E, to distinguish it from the Dirac matrices Γ_A.
- Mathematica names: `Γ17[A]`, `βD`, `αD[i]`, `ω17`, `χ17`, `EL`, `ER`, `Wch`, `ϕL`, `ϕR`, `σW[i]`, `dirac16complex`, `hKS`, `hChiKS`, and so on; in the notebook text cells the same symbols are written in plain Unicode.

4. Verification record

4.1 The Mathematica run

item	value
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)
method	`tools/run_eval.wl`: all cells evaluated in a fresh kernel (front-end kernel configuration "Eval71"; time limit 3600 s), then `NotebookSave`; requested through `tools/watch_requests.wl` (job `eval`)
finished	Tue 27 Oct 2026 14:15:35
evaluation time of the whole notebook	225.3 s
private context of the notebook	`Notebook$$41$124309`
input cells / output cells / message cells	145 / 382 / 0
checks in sections 7–9 / True	74 / 74
checks in the whole notebook / True	346 / 346
summary of section 10	`Length[checks]` = 202, `failedChecks` = {}, `allChecksPassed` = True
SHA-256 of the evaluated 7+1.nb (1034401 bytes)	`060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`

Before the final evaluation, every input cell of sections 7–9 was developed with `tools/eval_cells_log.wl`, which evaluates a generated notebook expression by expression in a fresh context, records the time, the messages and the result of every check, and stops at the first failure.

4.2 The checks of sections 7–9 of 7+1.nb and their results

Each row is one cell `check["name", test]` of the notebook, in order; the test is the Wolfram Language expression in that cell, and the result is what Wolfram 15.0.1 returned.

7.1 The eight real Dirac matrices and the Dirac α, β

#	check	result
1	`Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1`	True
2	`Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7`	True
3	`βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric`	True
4	`αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16`	True
5	`ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]`	True

7.2 The field dirac16complex and its Dirac equation

#	check	result
6	`(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)`	True

7.3 Hamiltonian form

#	check	result
7	`Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)`	True

7.4 Plane waves and the real spinors u_s, v_s

#	check	result
8	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
9	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
10	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
11	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
12	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
13	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
14	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True

7.5 Spin sums, completeness, and the diagonalization of h

#	check	result
15	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
16	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
17	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True

7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current

#	check	result
18	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
19	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
20	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
21	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True

7.7 Quantization: mode expansion and canonical anticommutators

#	check	result
22	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True

7.8 Charge conjugation

#	check	result
23	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
24	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True

7.9 Lorentz covariance

#	check	result
25	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True

7.10 Chirality: no real chirality matrix, complex chiral projections

#	check	result
26	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
27	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
28	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True
29	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True

7.11 The two coupled 8-component equations

#	check	result
30	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
31	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
32	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
33	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
34	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
35	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True

8.1 The self-coupled field: Lagrangian, densities, Hamiltonian

#	check	result
36	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
37	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True

8.2 The nonlinear field equation (Grassmann left derivatives)

#	check	result
38	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True

8.4 The 16-component Kohn–Sham equations

#	check	result
39	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
40	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
41	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True

8.5 Exchange: Wick's theorem for contact couplings

#	check	result
42	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True

8.6 The exchange self-energy and the Fierz identities

#	check	result
43	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
44	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
45	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
46	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True

8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock

#	check	result
47	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
48	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
49	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
50	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
51	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
52	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
53	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
54	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
55	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
56	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
57	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
58	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True

9.1 Chiral densities

#	check	result
59	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
60	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True

9.2 The Hartree energy in chiral variables

#	check	result
61	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

9.3 The chiral Kohn–Sham equations

#	check	result
62	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
63	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
64	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True

9.4 Chiral symmetry

#	check	result
65	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
66	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
67	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True

9.5 Uniform matter in chiral form

#	check	result
68	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
69	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True

9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

#	check	result
70	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
71	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True

#	check	result
72	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
73	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
74	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

4.3 Independent Python checks: tools/ref_dirac16.py

The program recomputes the constructions of sections 7–9 without Mathematica (Python 3.13.17, sympy 1.14.0, mpmath 1.3.0, numpy 2.5.3; its random test data come from the seed 71 of `random` and of numpy's `default_rng`). Its 90 boolean checks:

#	check	result
1	Gamma17 real, entries 0,+-1	True
2	{Gamma_A,Gamma_B} = 2 eta17_AB	True
3	Gamma0 symmetric, Gamma_i antisymmetric	True
4	Gamma^A dagger = Gamma0 Gamma^A Gamma0	True
5	alpha^i, beta real symmetric	True
6	{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1	True
7	block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]]	True
8	omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1	True
9	(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi	True
10	Gamma0 (i d0 - h) psi = (i Dslash - m) psi	True
11	h(p) = [[0,M],[M^T,0]]	True
12	h(p)^2 = E^2 1	True
13	M M^T = M^T M = E^2 1_8	True
14	h(p) u_s = E u_s, h(-p) v_s = -E v_s	True
15	(pslash - m) u_s = 0, (pslash + m) v_s = 0	True
16	u_s^T u_r = 2E delta, v_s^T v_r = 2E delta	True
17	u_s(p)^T v_r(p~) = 0	True
18	sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m	True
19	completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16	True
20	U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8)	True
21	charge conjugation: (pslash + m) omega u_s = 0	True
22	omega u_s = sum_r v_r (-M^T/E)_rs	True
23	S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost)	True
24	Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0	True
25	real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255)	True
26	omega anticommutes with all Gamma17[A]	True
27	chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0	True
28	[chi, Sigma_AB] = 0 (chirality is Lorentz invariant)	True
29	P_L, P_R not real	True
30	P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each	True
31	Gamma_A P_L = P_R Gamma_A, P_L^dagger Gamma0 = Gamma0 P_R	True
32	E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R	True
33	chi E_L = -E_L, chi E_R = E_R, W unitary	True
34	W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]]	True
35	sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta	True
36	chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag alpha E_R = sigma, E_L^dag alpha E_R = 0	True
37	E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R	True
38	E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L	True
39	(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components)	True
40	psi = E_L phi_L + E_R phi_R	True
41	chiral form of the bilinears (beta i chi, chi, alpha chi)	True
42	{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode algebra)	True
43	{psi_a(t,x), psi_b(t,y)} = 0	True
44	the three U_k solve the Dirac equation (m = 1)	True
45	Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0	True

#	check	result
46	current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field)	True
47	dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi) G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a	True
48	h_KS Hermitian for real S, P, V_A, A_A	True
49	Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p) (plane wave e^{-i(eps t - p.x)})	True
50	W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V + A, M = m + S + iP	True
51	Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A, A_A = gA j5_A	True
52	densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi	True
53	chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL	True
54	E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR	True
55	V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents)	True
56	M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP	True
57	dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i V^R_i) phi_R + (M - m)^* phi_L	True
58	U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R = e^{i theta} E_R	True
59	U^dag beta U = beta (cos 2theta + i sin 2theta chi); vector and axial currents invariant	True
60	E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2] -> U(1)_A symmetric iff gP = gS	True
61	256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T	True
62	tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256)	True
63	Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb	True
64	delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A R G^A - gA G_A chi R G^A chi	True
65	E_x = (1/2) tr[Sigma_x R]	True
66	E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR]	True
67	Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel vertices	True
68	uniform R: tr R = rho_S, tr(Gamma^0 R) = rho	True
69	(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E)	True
70	E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32 (8rS^2+6rho^2)	True
71	Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)	True
72	S_x = dE_x/drho_S, V_x = dE_x/drho	True
73	uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV + (-gS-gP+6gV+6gA)/16	True
74	uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0)	True
75	Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O' R) for all 18 vertex pairs	True
76	c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4)	True
77	Hartree gap: m* - m - gS rhoS(m*) = 0	True
78	m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2	True
79	total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H	True
80	Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho	True
81	d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger)	True
82	Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated)	True
83	HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x	True
84	HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V)	True
85	HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4	True
86	J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6	True
87	NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0	True
88	NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua)	True
89	NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0	True
90	sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E; tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E	True

The numbers it computes (30-digit arithmetic). `tools/verify_7p1_nb.py`, whose output is printed below together with that of `tools/ref_dirac16.py`, compares them with the numbers printed by 7+1.nb:

quantity	value
Hartree saturation kF0	0.75663615070149289469
Hartree rho0	0.00001388100213231687719 7
Hartree m*	0.34809734014624326514
Hartree rhoS	$6.5190265985375673486e-6$
Hartree V0	0.13881002132316877197
Hartree E/A - m	−0.028321427964738575854
Hartree eps_F	0.97167857203526142415
Hartree kF=1: m*	0.080964430277606126329
Hartree kF=1: E/A - m	0.41086484863433217115
Hartree kF=1: eps_F	1.9809848613081542876
HF couplings G_S, G_V	$(mpf('-98750.0'), mpf('20000.0'))$
HF kF=3/4: m*	0.36643630198154392124
HF kF=3/4: V0	0.26101775473161552570
HF kF=3/4: E/A - m	0.039146338016119861526
Hartree kF=3/4: m*	0.36297394764803019822
Hartree kF=3/4: E/A - m	−0.028065375846872960879
NJL g_c	−8766.8181930602193513
NJL g = 6/5 g_c	−10520.181831672263222
NJL |M| (m = 0)	0.55369385182924169122
NJL kappa = -4 M J(|M|) (M real)	−0.000026315792858364874200
NJL E(|M|) - E(0)	$-1.3102449629010022473e-6$
NJL M (m = 1/100)	0.58531284674622051123

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells
    ...: 145
output cells
    ...: 382
message cells (must be 0)
    ...: 0
check outputs
    ...: 310
check outputs that are True
    ...: 310
check["..."] calls in the input cells
    ...: 310
$Version output
    ...: ['{"15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$41$124309`"}']
private notebook context
    ...: True
Length[checks]
    ...: 346
failedChecks == {}
    ...: True
allChecksPassed output True
    ...: True
tau[A] printed == 4+4 tau[A] (python)
    ...: True
tauO[A] printed == python tauO[A]
    ...: True
Gamma17[A] printed == python G17[A]
    ...: True
gridO printed == python octonion table
    ...: True
gridx44 printed == python split table (4+4.nb)
    ...: True
Lambda boost/rotation printed
    ...: True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check)
    ...: True
classification table printed
```

```
    ...: True
section 7: check outputs (all True)
    ...: 35 (True)
section 8: check outputs (all True)
    ...: 23 (True)
section 9: check outputs (all True)
    ...: 16 (True)
section 7: printed Sqrt[5] u_s(p) == python (5 e_s; M^T e_s) and (pslash - m) u_s == 0
    ...: True
section 8: printed closed forms of int k^6 m*/E and int k^6 E: d/dkF == integrands, zero at kF
    ...= 0 (sympy)                                              : True
sections 8-9: 23 printed numbers agree with tools/ref_dirac16.py to 17 digits
    ...: True
sections 8-9: G_S == -98750, G_V == 20000 printed
    ...: True
tools/ref_dirac16.py: all its boolean checks True
    ...: True
sections 0-6: the 166 input and text cells equal those of the backup
    ...: True
section 11: check outputs (all True)
    ...: 7 (True)
section 12: check outputs (all True)
    ...: 25 (True)
section 13: check outputs (all True)
    ...: 15 (True)
section 14: check outputs (all True)
    ...: 14 (True)
section 15: check outputs (all True)
    ...: 16 (True)
section 16: check outputs (all True)
    ...: 8 (True)
section 17: check outputs (all True)
    ...: 10 (True)
section 18: check outputs (all True)
    ...: 7 (True)
section 19: check outputs (all True)
    ...: 6 (True)
summary table: entries == Length[checks], all True
    ...: True
summary table: every printed check is listed
    ...: True
sections 0-10: the 368 input and text cells equal those of the backup taken before sections
    ...11-20                                                    : True
sections 14-19: 27 groups of printed numbers agree with the Python programs (relative 1e-6;
    ...rounded prints 1e-4; ODE and ill-conditioned results 1e-5) : True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref_dirac16.py (independent checks of sections 7-9; exit code 0)

```
Gamma17 real, entries 0,+-1 : True
{Gamma_A,Gamma_B} = 2 eta17_AB : True
Gamma0 symmetric, Gamma_i antisymmetric : True
Gamma^A dagger = Gamma0 Gamma^A Gamma0 : True
alpha^i, beta real symmetric : True
{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1 : True
block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]] : True
omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1 : True
(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi : True
Gamma0 (i d0 - h) psi = (i Dslash - m) psi : True
h(p) = [[0,M],[M^T,0]] : True
h(p)^2 = E^2 1 : True
M M^T = M^T M = E^2 1_8 : True
```

```
h(p) u_s = E u_s,  h(-p) v_s = -E v_s : True
(pslash - m) u_s = 0, (pslash + m) v_s = 0 : True
u_s^T u_r = 2E delta, v_s^T v_r = 2E delta : True
u_s(p)^T v_r(p~) = 0 : True
sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m : True
completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16 : True
U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8) : True
charge conjugation: (pslash + m) omega u_s = 0 : True
omega u_s = sum_r v_r (-M^T/E)_rs : True
S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost) : True
Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0 : True
real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255) : True
omega anticommutes with all Gamma17[A] : True
chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0 : True
[chi, Sigma_AB] = 0 (chirality is Lorentz invariant) : True
P_L, P_R not real : True
P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each : True
Gamma_A P_L = P_R Gamma_A,  P_L^dagger Gamma0 = Gamma0 P_R : True
E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R : True
chi E_L = -E_L, chi E_R = E_R, W unitary : True
W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]] : True
sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta : True
chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag
    ...alpha E_R = sigma, E_L^dag alpha E_R = 0 : True
E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R : True
E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L : True
(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components) : True
psi = E_L phi_L + E_R phi_R : True
chiral form of the bilinears (beta i chi, chi, alpha chi) : True
{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode
    ...algebra) : True
{psi_a(t,x), psi_b(t,y)} = 0 : True
the three U_k solve the Dirac equation (m = 1) : True
Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0 : True
current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field) : True
dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi)
    ...G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a : True
h_KS Hermitian for real S, P, V_A, A_A : True
Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p)  (plane wave
    ...e^{-i(eps t - p.x)}) : True
W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V
    ...+ A, M = m + S + iP : True
Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A,
    ...A_A = gA j5_A : True
densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi : True
chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL
    ...: True
E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR : True
V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents) : True
M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP : True
dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i
    ...V^R_i) phi_R + (M - m)^* phi_L : True
U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R =
    ...e^{i theta} E_R : True
U^dag beta U = beta (cos 2theta + i sin 2theta chi);  vector and axial currents invariant : True
E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2]
    ...->  U(1)_A symmetric iff gP = gS : True
256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T : True
tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256) : True
Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb : True
delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A
    ...R G^A - gA G_A chi R G^A chi : True
E_x = (1/2) tr[Sigma_x R] : True
E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR] : True
Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel
```

```
    ...vertices : True
uniform R: tr R = rho_S, tr(Gamma^0 R) = rho : True
(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E) : True
E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32
    ...(8rS^2+6rho^2) : True
Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)
    ...: True
S_x = dE_x/drho_S, V_x = dE_x/drho : True
uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV +
    ...(-gS-gP+6gV+6gA)/16 : True
uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0) : True
Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O'
    ...R) for all 18 vertex pairs : True
c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4) : True
NUM Hartree saturation kF0 : 0.75663615070149289469
NUM Hartree rho0 : 0.00001388100213231687719 7
NUM Hartree m* : 0.34809734014624326514
NUM Hartree rhoS : 6.5190265985375673486e-6
NUM Hartree V0 : 0.13881002132316877197
NUM Hartree E/A - m : -0.028321427964738575854
NUM Hartree eps_F : 0.97167857203526142415
Hartree gap: m* - m - gS rhoS(m*) = 0 : True
m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2 : True
total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H : True
Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho : True
d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger) : True
Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated) : True
NUM Hartree kF=1: m* : 0.080964430277606126329
NUM Hartree kF=1: E/A - m : 0.41086484863433217115
NUM Hartree kF=1: eps_F : 1.9809848613081542876
NUM HF couplings G_S, G_V : -98750.0, 20000.0
HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x :
    ...True
HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V) : True
HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4 : True
NUM HF kF=3/4: m* : 0.36643630198154392124
NUM HF kF=3/4: V0 : 0.2610177547316155257
NUM HF kF=3/4: E/A - m : 0.039146338016119861526
NUM Hartree kF=3/4: m* : 0.36297394764803019822
NUM Hartree kF=3/4: E/A - m : -0.028065375846872960879
J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6 : True
NUM NJL g_c : -8766.8181930602193513
NUM NJL g = 6/5 g_c : -10520.181831672263222
NUM NJL |M| (m = 0) : 0.55369385182924169122
NUM NJL kappa = -4 M J(|M|) (M real) : -0.00002631579285836487 42
NUM NJL E(|M|) - E(0) : -1.3102449629010022473e-6
NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0 :
    ...True
NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua) : True
NUM NJL M (m = 1/100) : 0.58531284674622051123
NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0 :
    ...True
sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E;
    ...tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E : True
boolean checks: 90   failed: 0
VERDICT: ALL CHECKS PASSED
```

4.4 Hashes of the files

file	SHA-256
7+1.nb (evaluated)	060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
7+1_2026-10-26_before_dirac16complex.nb (in backups/, workspace only)	f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	eb63a44f3da35fc75dae8420186867f85fecfa15415eefa108b5b0f23c646385
tools/doc_dirac16.py	7916c204596a471015e9408616aaca1b7b89575adc774b808c468f337e8418de
tools/doc_dirac16_common.py	9324b4fae59e51019125ebe09157f3c8c9489560e1383a697fe6c22a3a93e208
tools/doc_dirac16_provenance.py	9ac843eac503c8f3d1417a67e82765c0f896422171a897a3200b67f80d420dfe
tools/doc_dirac16_symbols.py	70cee4b84501c0f7a7a02105e39acfa6e21c61835a667eca28e43e6ad65cf674
tools/doc_gravity.py	6d0da11131b4f0473855658cdb1c2a34afbdac630cfcf528a9ecbe5cf5ef3d90
tools/doc_gravity_provenance.py	2809bf6cc4f6c44c4f9a66422321329103bb4d231cbc3c1e7eea794026fe77f4
tools/doc_gravity_symbols.py	da852b5feb4a7d68b44da53eb254471d1c8546d305b6dda3c8e59440e1372ae8
tools/doc_ks_gravity.py	e62dffaadee563428788e9600ba14c92acc346186d58d70a9b41193e917f9ff8
tools/doc_ks_gravity_provenance.py	1a97d4be904ff2921c426ae92dee4ef3a3eb9dde20f2f9ef55babb5a9923e4f1
tools/doc_ks_gravity_symbols.py	afa9b5b30b13e5ab90a2cf74bd6ae5ef8a5dbaa34c25ce33a1ce6011033f9fbd
tools/doc_provenance.py	49e7b60c6ce31ad7869b3b9c3eaefd28517e8b4ffd7e4d8c0e9651d461bb56a1
tools/doc_readme.py	1b99079a18c39f9ba7a73465982787c9b9139079d3a4af98e65a261bb5eaacff
tools/doc_readme_stage3.py	8076bbaf4f0c8854a9411f95cd1637fc1dcfeae1fc45f2701d88ff7f356a67d6
tools/doc_sm.py	faa244fed5ec09465f576661ad601def9e815e56389361e037e98fd5b45553c0
tools/doc_sm_provenance.py	ab0f9d1cce27f2250bfa278ff72670bd40339cfa8ca0ecc025c3e099a2d4118a
tools/doc_sm_symbols.py	353df4989623fed65925777ce46759ff296f3cec3bd42797123b7b4b0b3d9766
tools/doc_stage3_common.py	2364ce6eff9533051a9b56a5435e59039cf9417c03a477d75d40f68c8dd4b1d6
tools/doc_symbols.py	f5bf59d7962459f7e8bc06581852e0be3b695f7cec7513b01a9e91fcfa1fa507
tools/docgen.py	5db6f40cb1eb10e48da5b69c3893829737c3d8d2e90edcb28311a980c1aa5fea
tools/eval_cells_log.wl	833b6bd854cab253e5bb935c28f28ab50b286f409d130f433fa5dfb9b9236bbe
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/git_push.wl	ac05a6d0256834b48cc9fa943fa247bccc574f723e68f053d00bfba9aca2e21a
tools/lovelock71.py	d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af95a1
tools/make_7p1_nb.py	2fac94dfd0c9fcf3aed8d6e90a77fbbb8ab1c8450819ff591b7e5cf7b5787d6f
tools/make_docs.py	2c71a87dc93d5612106129a671af4f67b351d4f290e586388f42106a39b74f81
tools/nb_part_gravity.py	55d16f735e16df7a7cd2aa70d322969bc6cf82c777965f112b5a9acf46a6ad5f
tools/nb_part_ks_gravity.py	886814fedc4cd015c63b950ee83724025ba21fd6c3e56b68dcb7f2f07afdc205
tools/nb_part_sm.py	20782f1fe557fd3f3dea7417e89ecd42a1602f7a48599fcdcfd0cff77f120ecc
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5
tools/ref71.py	4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb8b0
tools/ref_dirac16.py	a6e139ebe5199621a95c8adc772bc3e1e9b452c0ca20a963c536cbfc6539319c
tools/ref_gravity.py	2bfdc087ed4dc7d84258e04449d11141b415f54b992e65d42fa42cace08ff6bd
tools/ref_ks_gravity.py	8d082985c5161254d7c6294c0310f7807aa037bff27a8cd3ec4d3f8a9f6e0550
tools/ref_sm.py	d36d23f472c8339d80f1d0b1eb46c509398562053ead7033d96b054782e73b49
tools/run_eval.wl	85ba54f18a8362a69c991e66a207230dd001ba18e52639fc1edf7cac761aac30
tools/verify_7p1_nb.py	d3aaeeeb7f380f63d4e7610c5b4235973cadd1d6db0ebf09ecbd65d561795d17
tools/watch_requests.wl	d9ceedb0b45ac3cf9b4dd8519ac164953f2c80e5188b1557fe20ef5107ae352d

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows. Python and LaTeX ran in a Linux shell with that folder mounted; intermediate files went to `build/`, which the repository's .gitignore excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (sections 0–6 by the code of the first stage, unchanged; sections 7–10 appended), with cells written by `tools/nbwrite.py` as pure ASCII (named characters such as `\[CapitalGamma]`, and `\:03d6` for ϖ).
- All Wolfram Language work ran in the Wolfram 15.0.1 front end (the program of section 4.1), version 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026). Jobs were requested without keyboard input through `tools/watch_requests.wl`: loaded once in that front end, it reads `build/request.txt` every 10 seconds and evaluates `tools/run_eval.wl` (job `eval`), `tools/git_push.wl` (job `push`) or a development file such as `tools/eval_cells_log.wl` (job `run`).
- Python 3.13.17 with sympy 1.14.0, mpmath 1.3.0 and numpy 2.5.3; fontTools reads the font coverage for the LaTeX output.
- This record and the main document and symbol dictionary of this stage are generated by `tools/make_docs.py` from the modules `docgen.py`, `doc_common.py`, `doc_dirac16_common.py`, `doc_dirac16.py`, `doc_dirac16_symbols.py` and `doc_dirac16_provenance.py` of the folder tools/; one source renders Markdown and LaTeX, and every matrix, check list and number is computed by Python or parsed from the evaluated 7+1.nb. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu).
- Pushes: `tools/git_push.wl` runs the user's Git for Windows from the Wolfram front end with `GIT_TERMINAL_PROMPT=0` and `credential.interactive=never` (no login prompt), pushes the branch main to origin `https://github.com/once-ere/7plus1.git`, and logs `git ls-remote origin`; the published commit is compared with the local one after every push.
- Order of the work: backup of 7+1.nb; Python reference `tools/ref_dirac16.py`; notebook sections 7–10, developed cell by cell; evaluation and cross-check (`tools/verify_7p1_nb.py`); documentation. The notation of the complex scalar source was changed from μ to ϖ after the first evaluation, and 7+1.nb was regenerated and evaluated again; the boost, the proof matrices and Euler's Gamma function were renamed in the documents only ($\mathcal{B}$, Y, Z, Γ_{E}).

6. Authorship

Notebook sections 7–10, tools and documents written by Claude (Anthropic, model claude-opus-5-5) for Nash, completed on 2026-10-27, from the task described in section 1 and the first part of 7+1.nb described in section 2.1.